

STATS 507 Final Project Proposal

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November 25, 2024

1 Overview

1.1 background and motivation

Diabetes is a chronic condition that affects millions of people worldwide, and its early detection is crucial for preventing serious complications. Predicting whether a person has diabetes can significantly improve healthcare outcomes by enabling timely interventions. With the increasing adoption of machine learning, predictive models have become valuable tools in healthcare for diagnosing conditions like diabetes. My primary motivation for this project is to explore the application of machine learning in healthcare, specifically for predicting diabetes. By building and evaluating models, I aim to gain insights into the model development process and understand how machine learning can be leveraged in healthcare. Through this project, I also hope to gain hands-on experience comparing both simple and advanced models, allowing me to evaluate how different algorithms perform for the same task and ultimately contribute to improving diabetes prediction.

1.2 data set and models

For this project, I will work with the Pima Indians Diabetes Dataset, which includes health-related data such as glucose levels, BMI, age, and other diagnostic features for 768 female patients. The target variable indicates whether a patient has diabetes (1) or not (0). I will start with Logistic Regression as the baseline model, which is known for its simplicity and interpretability. After assessing the performance of the logistic regression model, I plan to explore more advanced models, such as XGBoost, Random Forest, and CNNs, to see if they provide better accuracy or improved predictions.

1.3 Expected Insights

I expect to uncover valuable insights regarding which features are most predictive of diabetes, and how the performance of different models compares. By

evaluating the performance of logistic regression, I hope to gain a deeper understanding of its strengths and limitations. Furthermore, I aim to explore advanced models and determine if they lead to improved predictive performance, particularly in terms of accuracy, recall, precision, and F1-score.

2 Prior Work

2.1 literature review

In recent years, machine learning has become a powerful tool in healthcare, particularly for predicting diseases like diabetes. Logistic Regression, known for its simplicity and interpretability, is commonly used for binary classification tasks such as diabetes prediction based on clinical features like age, BMI, glucose levels, and blood pressure. While studies like Badu et al. (2020) highlight its effectiveness, more advanced models such as Random Forests (RF) and XGBoost have shown superior performance. RF, with its ensemble approach, reduces over-fitting and handles high-dimensional data well (Breiman, 2001), while XGBoost has become a go-to method for capturing complex patterns, improving accuracy, and reducing errors, as demonstrated by Ganaie et al. (2021) and Gundogdu et al. (2019). Though deep learning methods like Convolutional Neural Networks (CNNs) and Artificial Neural Networks (ANNs) have shown promise in complex tasks, they generally require larger datasets and more computational resources.

2.2 potential methods

I plan to begin with Logistic Regression to establish a baseline and then explore more advanced models like XGBoost, RF, and possibly CNNs to see if they offer better performance in diabetes prediction.

3 Preliminary Results

3.1 data understanding

The project will use the Pima Indians Diabetes dataset, which contains 768 samples with 8 clinical features such as age, BMI, glucose levels, and blood pressure, along with a binary target variable indicating whether the patient has diabetes. An initial assessment of the dataset will focus on checking for missing values, data distributions, and potential outliers. The dataset is relatively clean, but challenges may arise from handling skewed or imbalanced data, which could affect model performance.

3.2 EDA plots

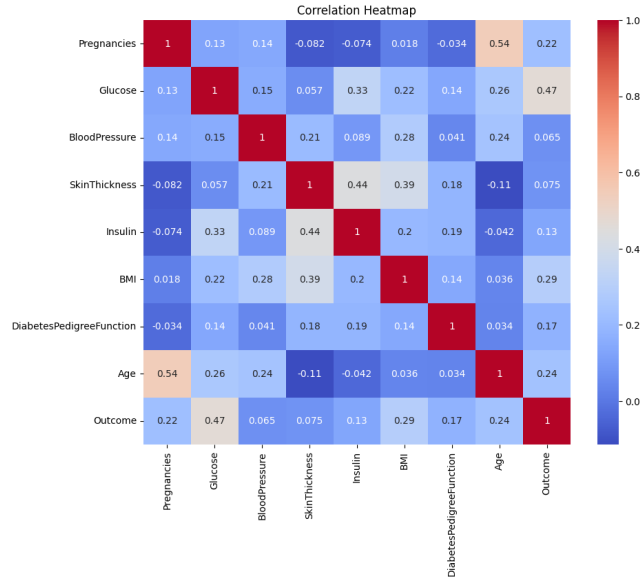


Figure 1: Correlation Heatmap of Features

Heatmap (Correlation Matrix): The heatmap visualizes the correlations between numerical features in the dataset. By using color gradients, it highlights the strength and direction of relationships between pairs of features. This helps identify which features are strongly correlated, guiding decisions on potential feature engineering or handling multicollinearity during modeling.

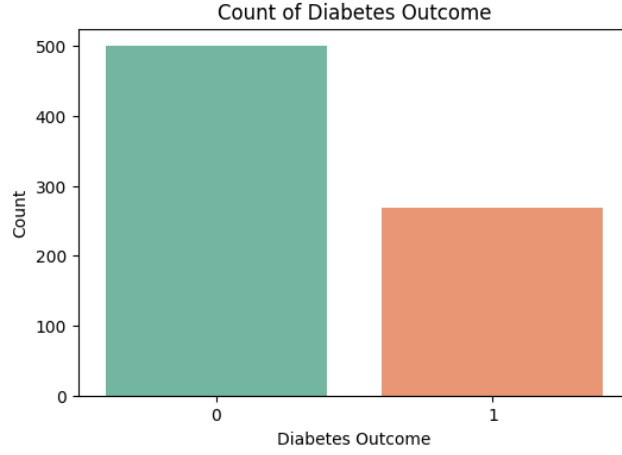


Figure 2: Countplot of Outcome Variable (Diabetes vs. No Diabetes)

Countplot (Target Variable Distribution): The countplot visualizes the distribution of the target variable, Outcome, which indicates whether a person has diabetes (1) or not (0). This plot helps assess the class balance in the dataset, which is important for understanding the impact on model performance and whether any class imbalance techniques might be necessary.

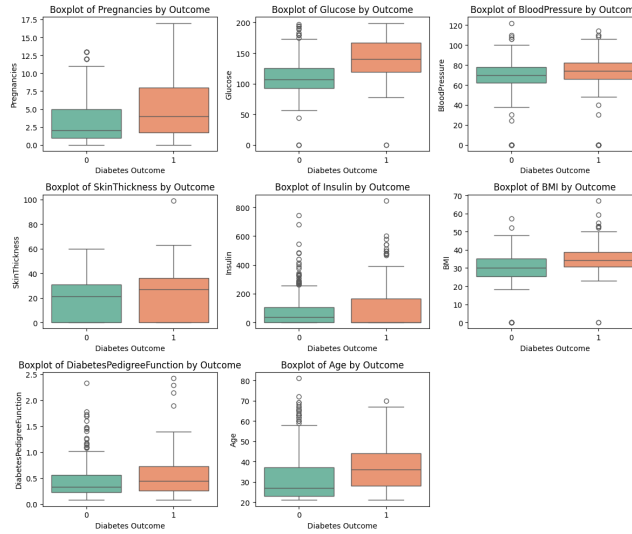


Figure 3: Boxplot of Features by Outcome

The boxplot displays the distribution of each feature across the two outcome classes (diabetic vs. non-diabetic). It shows the median, quartiles, and poten-

tial outliers for each feature, helping identify how feature values differ between the two classes and whether any features might be particularly important for distinguishing between them.

3.3 basic model

For the basic model, I will start by implementing Logistic Regression as a baseline to predict whether a patient has diabetes. The resource requirements for Logistic Regression are minimal, as it is a computationally efficient model. However, performance bottlenecks may arise when trying to improve accuracy through feature scaling or hyperparameter tuning. This will be addressed by iterative model refinement and validation.

The logistic regression model for binary classification predicts the probability of an outcome being 1 (e.g., the presence of diabetes) given a set of input features. The model is based on the logistic equation, which applies the sigmoid function to a linear combination of the input features. Specifically, the probability $P(Y = 1 | X)$ is modeled as:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_8 X_8)}}$$

where β_0 is the intercept and $\beta_1, \beta_2, \dots, \beta_8$ are the coefficients of the predictor variables. The output of this function is always between 0 and 1, representing the probability of the positive class. Based on this probability, the model classifies the outcome as 1 if $P(Y = 1 | X) \geq 0.5$, and as 0 otherwise.

3.4 tools

For this project, I will be using several tools covered in class as well as some additional libraries for advanced exploration. From class, I will be utilizing:

- **Scikit-learn:** For implementing the Logistic Regression model and evaluating its performance.
- **Pandas:** For data manipulation, cleaning, and preparing the dataset.
- **NumPy:** For numerical operations and handling arrays.
- **Seaborn/Matplotlib:** For visualizing the dataset and model performance.

Additionally, I plan to explore the following tools outside of class to enhance the project:

- **XGBoost:** For implementing a more advanced ensemble model to compare with Logistic Regression.
- **TensorFlow/Keras:** For potential future exploration of deep learning models, if required for complex tasks.

4 Project Deliverables

A successful project will produce a predictive model capable of determining whether a person has diabetes based on clinical data. The final deliverables will include a detailed comparison of Logistic Regression, XGBoost, and possibly other advanced models, including performance metrics such as accuracy, precision, recall, and F1-score. The project will also include a comprehensive analysis of model improvements, feature selection, and adjustments made throughout the process.

sub-goals

- Build and evaluate the Logistic Regression model as the baseline.
- Implement and compare more advanced models (XGBoost, Random Forest, etc.).
- Assess the performance of different models and make adjustments to improve accuracy.

5 Timeline

- **Week 1-2:** Data exploration, understanding dataset statistics, cleaning data, and building the Logistic Regression model.
- **Week 3-4:** Evaluate the Logistic Regression model, adjust features, and implement XGBoost and other advanced models.
- **Week 5:** Compare model performances, finalize the best approach, and prepare project documentation for submission.